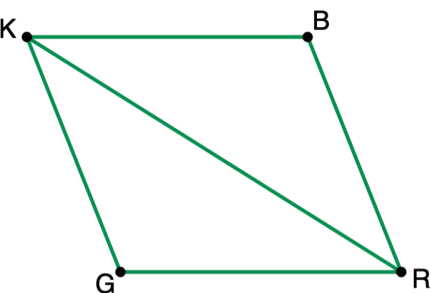




5.5.6 Wrap-Up: Ticket Out the Door

Quadrilateral $BRGK$ is shown.



Complete each statement based on the given information.

1. Given that $\overline{KB} \cong \overline{GR}$ and $\overline{KG} \cong \overline{BR}$,
 - a. I can conclude that ...
 - b. I know this because ...
2. Given that $\overline{KB} \cong \overline{KG}$ and $\overline{RG} \cong \overline{BR}$,
 - a. I can conclude that ...
 - b. I know this because ...
3. Given that $\overline{KG} \parallel \overline{BR}$ and $\overline{KB} \parallel \overline{GR}$,
 - a. I can conclude that ...
 - b. I know this because ...

Unit 5, Lesson 6: Proving Triangles Congruent



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Target

- ☐ I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.



5.6.1 Warm-Up: Cheese Balls

The image shows cheese balls arranged on a baking sheet with a white plate on top. There are thirteen cheese balls on the plate.



1. Estimate how many cheese balls will fit on the plate.
2. Explain how you determined your estimate.



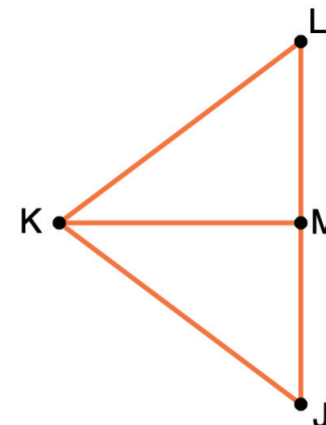
5.6.2 Guided Instruction: Proving Triangles Congruent

When completing a proof to show two triangles are congruent, it is helpful to start by considering conclusions that can be drawn from the given information.

1. Triangle KLJ is shown with some given information.

Given: $\overline{KM}$ bisects $\overline{LJ}$
 $\overline{KL} \cong \overline{KJ}$

- a. What conclusion can be drawn from the given statement, " $\overline{KM}$ bisects $\overline{LJ}$ "?
- b. Explain why this conclusion can be drawn.
- c. Mark the figure based on the given information.
- d. What other conclusion(s) can be drawn from the information given?

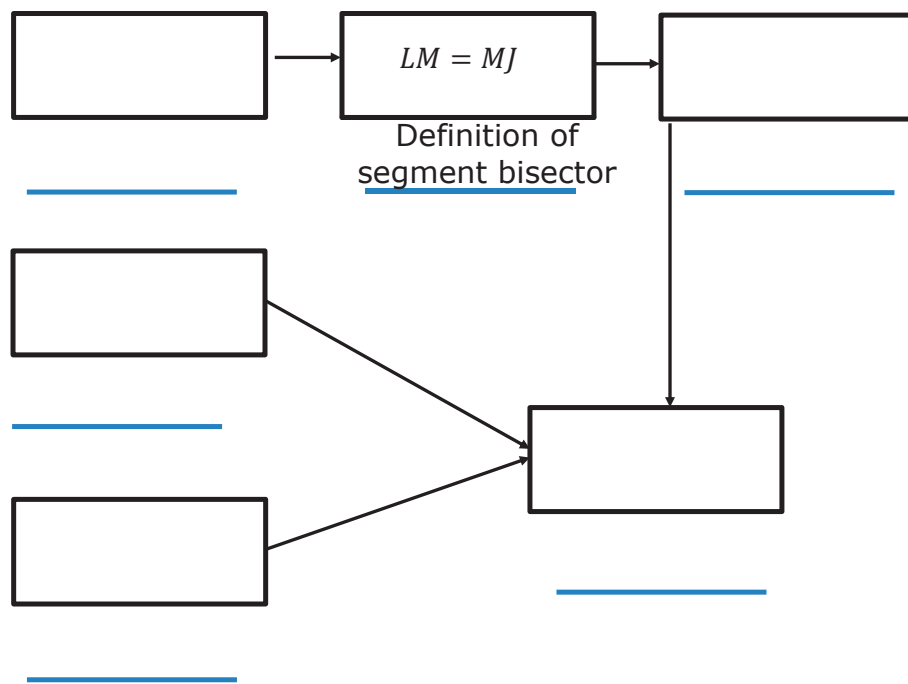


e. Explain why this conclusion can be drawn.

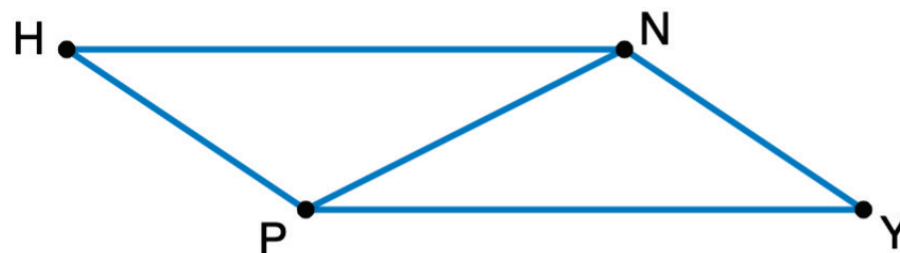
f. Use the conclusions drawn about $\triangle KIJ$ from the information given, along with the figure, to complete the flow chart proof.

Given: $\overline{KM}$ bisects $\overline{IJ}$
 $\overline{KL} \cong \overline{KJ}$

Prove: $\triangle KIM \cong \triangle KJM$



2. Polygon $HPYN$ is shown.



a. Mark the diagram to show the following given information.

Given: $\overline{HP} \cong \overline{YN}$ and $\overline{HP} \parallel \overline{YN}$

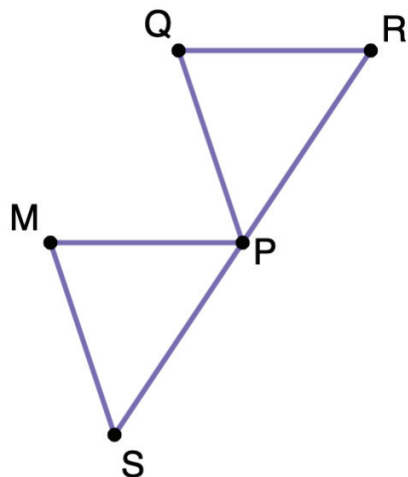
b. Discuss how you determined which markings to make with your partner.

c. Which congruence theorem or postulate should be used to prove $\triangle PHN \cong \triangle NYP$?

d. Complete the two-column proof.

Statement	Reason
1. $\overline{HP} \cong \overline{YN}$	1. Given
2. $\overline{HP} \parallel \overline{YN}$	2.
3.	3.
4.	4.
5. $\triangle PHN \cong \triangle NYP$	5.

3. Triangles MSP and QPR are shown.



Given: $\overline{QR} \parallel \overline{MP}$
 $\angle SMP \cong \angle PQR$
 P is the midpoint of $\overline{SR}$

Prove: $\triangle SPM \cong \triangle PRQ$

- Mark the diagram to show the given information.
- Complete the paragraph proof.

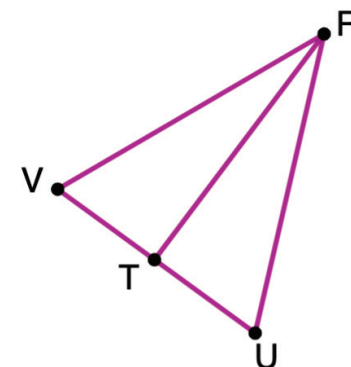
It is given that P is the midpoint of $\overline{SR}$, so _____
 by _____. It is given that $\overline{QR} \parallel \overline{MP}$,
 so $\angle _____ \cong \angle _____$ by _____.
 It is also given that _____. Therefore,
 $\triangle SPM \cong \triangle PRQ$ by _____.



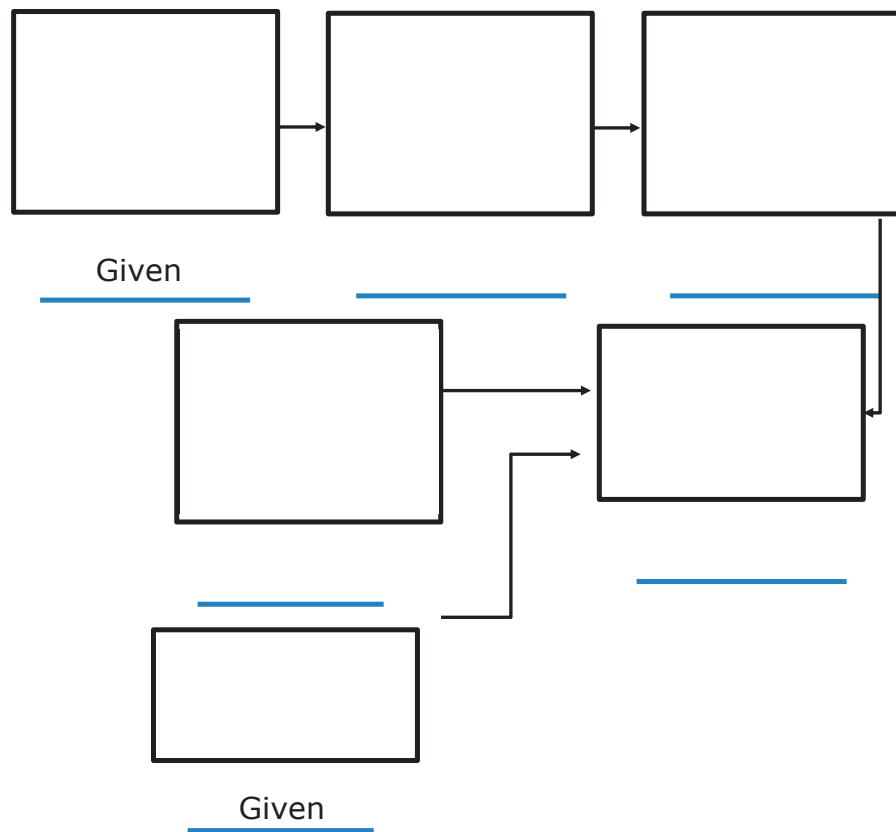
5.6.3 Guided Practice: Proving Triangles Congruent

1. Triangle VRU is shown.

Given: $\overline{UR} \cong \overline{VR}$
 $\overline{RT} \perp \overline{UV}$
 Prove: $\triangle RTV \cong \triangle RTU$



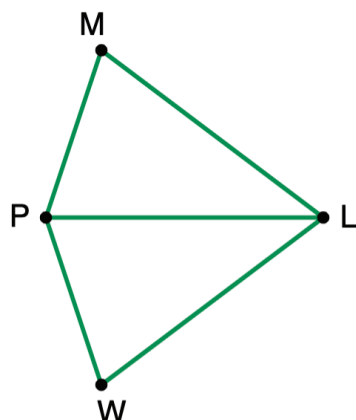
Complete the flow chart proof.





5.6.4 Your Turn!

1. Polygon $MLWP$ is shown.



Given: $\overline{LP}$ bisects $\angle MLW$
 $\angle LPM \cong \angle LPW$

Prove: $\triangle WPL \cong \triangle MPL$

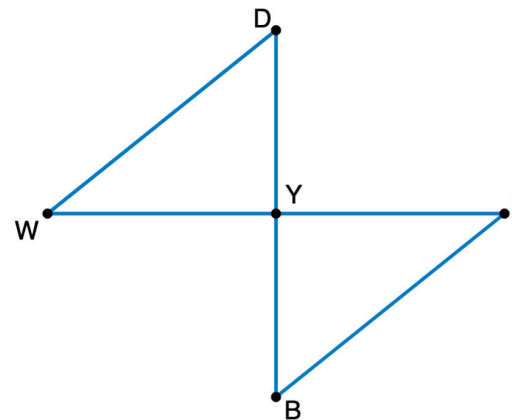
Complete the paragraph proof.

It is given that $\overline{LP}$ bisects $\angle MLW$, so _____ by definition of _____. Also, it is given that _____. _____ $\cong$ _____ by _____. Therefore, $\triangle WPL \cong \triangle MPL$ by _____.



5.6.5 Wrap-Up: Error Analysis

1. The flow chart proof has two errors. Find the errors and correct them.



Given: Y is the midpoint of $\overline{WL}$
 $\overline{DY} \cong \overline{BY}$

Prove: $\triangle DWY \cong \triangle BLY$

